# KEYSTONE: Readiness Audits for Dispute-Key Availability in Encrypted Rollups

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**Abstract**

Encrypted optimistic rollups can keep ciphertexts available and still fail a dispute because the threshold service that must decrypt the disputed record is unavailable, late or uncooperative. We call the missing property dispute-key availability (DKA) and separate it from five weaker availability properties. KEYSTONE audits readiness with sampled canary shares carrying Chaum–Pedersen proofs; their transcripts are simulatable from public shares, so audits exercise real shares without revealing records. We give exact operating characteristics and show that a single small audit cannot separate a committee one share short of its threshold from a live one, whereas sequential tests can. Under a stated assumption, record-labeled TDH2 restores the confidentiality plain threshold ElGamal loses to parties obtaining other releases. For custodians that fail and recover, a hidden-Markov recursion gives the exact failure risk after any audit history; without repair, auditing more often cannot beat a floor, and with resharing on alarm, sample size trades audits against mostly benign resharings. Exact analysis, simulation and a 32-process testbed confirm these results and show that selective withholders pass every audit yet defeat every dispute; in a replication specified in advance, an order-statistic deadline met its target except where no network delay absorbed host overhead.

**Keywords**

Dispute-key availability, threshold decryption, TDH2, encrypted rollups, readiness audit, acceptance sampling, hidden Markov model, selective withholding

## 1 Introduction

Optimistic rollups execute transactions off chain and accept the result unless a challenger proves it wrong within a dispute window[1]. When the executed workload is private, as in encrypted machine-learning inference[2,3], the records a challenger needs are published only as ciphertexts. Data-availability mechanisms can keep these ciphertexts retrievable. They cannot ensure that the committee holding shares of the decryption key delivers enough valid decryption contributions once a dispute is authorized and before its deadline. A ciphertext that cannot be opened in time is, for the dispute, unavailable. AI-rollup disputes are our running example, but the question arises wherever a protocol must open threshold-encrypted inputs on demand.

The failure is silent until it is needed. Custodians may lose shares, fail together when a provider or region goes down, or run broken software, and the key must stay serviceable for a dispute window that lasts days and may outlive the committee that sealed the record. A strategic custodian may also answer every routine check and refuse one targeted dispute; no audit of the kind studied here can detect that, and we treat it as a limit. Threshold encryption, verifiable and proactive secret sharing, and proofs of retrievability and data-availability sampling [4–9] answer adjacent questions; none produces public evidence that enough custodians can serve an authorized dispute now, or states what such evidence implies.

We study this property, *dispute-key availability* (DKA), and design and evaluate KEYSTONE, a readiness audit for it that targets non-strategic faults. KEYSTONE seals each record under a fresh key wrapped by a threshold Diffie–Hellman KEM. Routine audits ask a beacon-sampled subset of custodians to apply their shares to a *canary*, a group element hashed from public audit context, and to prove correctness with a Chaum–Pedersen equality-of-discrete-logarithm (DLEQ) proof; each response is structurally a threshold-coin or threshold-signature share on a hashed message [10,11]. An authorized dispute applies the same share operation to the disputed record. Three results matter most for deployments: a small audit cannot tell a committee one share short of its threshold from a live one; no audit frequency pushes the probability of an undetected failure below a floor set by the audit design; and no audit of this kind detects a custodian that withholds only targeted disputes. Our contributions are:

1. **A property decomposition** (Section 4) of $(\varepsilon, \Delta_d)$-DKA into five availability properties (ciphertext availability, valid-share registration, audit-time readiness, authorized decryptability and deadline liveness) and pre-authorization confidentiality, with witnesses for three inferences a deployment must not draw.
2. **A confidentiality-preserving audit** (Section 5) whose canary transcripts are simulatable from public shares, so audits exercise real shares without opening records. The plain threshold KEM is malleable across records; record-labeled TDH2 restores confidentiality against parties who obtain other releases.
3. **An exact analysis** (Section 6): operating characteristics, showing that small audits cannot separate a committee one share short of $t$ from a live one; sample sizes and sequential tests that certify a safety margin; a placement condition; a deadline budget; for custodians that fail and recover, the exact posterior risk given an audit history, a frequency floor, the longest audit interval meeting a risk target and the cost of resharing on alarm; and why targeted withholding escapes every pre-dispute test, with the stake that deters it.
4. **An evaluation with labeled evidence** (Section 7): exact analysis, prespecified simulation, microbenchmarks and a 32-custodian testbed with a replication specified in advance, with negative results: a high benign alarm rate under the strictest rule, a selective-withholding counterexample, deadlines missed without network delay and slower batch verification.

None of the primitives is new; the contribution is what audits built from them can and cannot certify, and a passed audit does not prove future availability.

Sections 2–7 give the model, properties, protocol, analysis and evaluation, and Sections 8–11 discuss them.

## 2 Preliminaries

*Groups and sharing.* Let $\mathbb{G}$ be a cyclic group of prime order $\ell$ with generator $g$ in which the decisional Diffie–Hellman (DDH) problem is hard. The prototype uses ristretto255, a prime-order group built on Curve25519[12], and keeps the quadratic residues modulo the RFC 7919 primes[13] (ffdhe2048, ffdhe3072; $\ell = (p-1)/2$, $g = 4$) as a baseline; we write groups multiplicatively, and $q$ denotes only the acceptance threshold of an audit, and $\binom{a}{b} = 0$ when $b > a$ or $b < 0$. A secret $x \in \mathbb{Z}_\ell$ is Shamir-shared with a polynomial $f$ of degree $t-1$, $f(0) = x$; custodian $i \in \{1, \dots, n\}$ holds $x_i = f(i)$ and publishes $Y_i = g^{x_i}$[14], verifiable against Feldman commitments[5]. Any $t$ distinct indices $I$ have Lagrange coefficients $\lambda_i^I$ with $\sum_{i \in I} \lambda_i^I x_i = x$.

*Threshold Diffie–Hellman KEM.* With epoch key $PK = g^x$, a sender draws $\rho \leftarrow \mathbb{Z}_\ell$, publishes $C_1 = g^\rho$ and derives a key-encryption key from $PK^\rho$ and the record identifier. Any $t$ partial decryptions $C_1^{x_i}$ determine $C_1^x = \prod_{i \in I} (C_1^{x_i})^{\lambda_i^I}$. This threshold ElGamal KEM[4] is semantically secure under DDH in the random-oracle model but malleable: anyone can turn $C_1$ into $C_1 g^\gamma$. Shoup and Gennaro's TDH2 binds a label and a proof of knowledge of $\rho$ to the ciphertext, which makes it secure against chosen-ciphertext attacks[15]; KEYSTONE uses it for disputes (Section 5.2).

*DLEQ proofs, hashing and beacon.* A Chaum–Pedersen proof[16] shows $\log_g Y_i = \log_X D_i$: the prover picks $\nu \leftarrow \mathbb{Z}_\ell$, sends $a_1 = g^\nu$, $a_2 = X^\nu$, derives $c = H(g, Y_i, X, D_i, a_1, a_2, \mathsf{ctx})$ by the Fiat–Shamir transform[17] and returns $z = \nu + c\, x_i$; the verifier checks $g^z = a_1 Y_i^c$ and $X^z = a_2 D_i^c$. The request context $\mathsf{ctx}$ ties each proof to one request. $H_G$ maps byte strings to $\mathbb{G} \setminus \{1\}$ with domain separation; we model $H$ and $H_G$ as random oracles. A public beacon outputs values that are unpredictable before publication[18].

## 3 System and threat model

An epoch $E$ has $n$ custodians placed in failure domains (providers, regions, software stacks), a threshold $t$, a key $PK_E$ and registered public shares $Y_1, \dots, Y_n$. A rollup worker encrypts each record $r$ under a fresh data key $K_r$ with AES-GCM, wraps $K_r$ under the KEM, and publishes the result to a data-availability (DA) layer. A settlement layer, the *bulletin board*, records registrations, requests, deadlines and evidence and supplies an ordered public clock. A *combiner* inside the authorized destination collects partial decryptions over authenticated, confidential channels; it learns $K_r$ for records it may open and is trusted not to disclose it. The board stores only commitments to responses with their validity status and timestamps, never enough to decrypt a record; canary responses may be public (Lemma 2). Figure 1 shows the three paths.

**Assumption 1.** Threshold confidentiality. During an epoch the adversary learns fewer than $t$ current shares. DDH is hard in $\mathbb{G}$; $H$, $H_G$ and the key-derivation function are independent, domain-separated random oracles; and AES-GCM is a secure authenticated-encryption scheme.

**Assumption 2.** Beacon. An audit's beacon value is unpredictable until its epoch, slot and committee are committed on the bulletin board, and the derived sample is uniform (without replacement, or within each domain quota for stratified samples) and independent of other

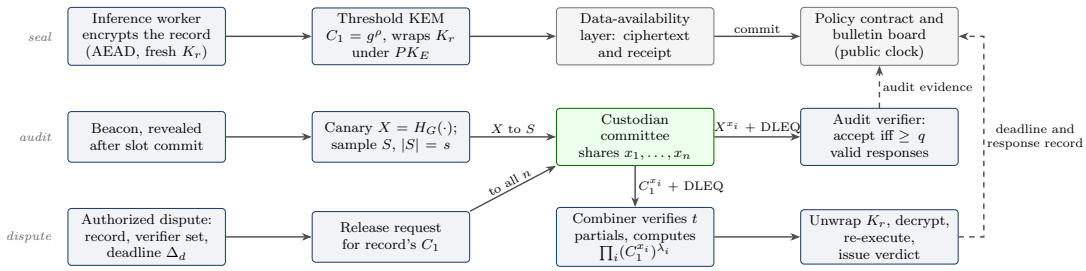



**Figure 1.** KEYSTONE architecture. Sealing, routine canary auditing and authorized dispute release are separate paths over one custodian committee. The audit uses a beacon-derived canary, never a production ciphertext; only an authorized dispute requests partial decryptions of a record's KEM component $C_1$. Dashed arrows are public evidence written to the bulletin board.

audits' samples. The prototype derives the sample with a SHA-256 counter stream and rejection sampling, which we model as ideal.

**Assumption 3.** Bulletin board and data availability. The bulletin board is append-only and totally ordered and exposes each request to custodians within $\delta_b$. The DA layer keeps committed ciphertexts and receipts retrievable, and record identifiers are unique.

**Assumption 4.** Bounded delay, for deadline statements only. Messages between non-faulty parties arrive within $\delta$ over channels that authenticate the sending custodian. A non-faulty custodian produces a partial decryption with proof within $c_p$; the combiner verifies one proof within $c_v$ and combines $t$ partials within $c_c$.

Detection and confidentiality statements do not use Assumption 4. The adversary may crash custodians individually or by domain, cause share loss, send invalid or equivocal responses, issue unauthorized requests, act as a worker that commits records of its choice and obtains their releases, and control *selective* custodians that answer canaries but withhold targeted dispute responses. Corruption is static within an epoch. Out of scope are production key-generation security, combiners or verifiers that leak records they may open, arbitrary asynchronous partitions, and selective withholding beyond Observation 1. An audit shows that whoever answered knows $x_i$, not that the registered operator does, so share pooling and false domain declarations are invisible to it; and threshold custody replaces the one-honest-challenger assumption of optimistic rollups with an assumption on $t$ of $n$ custodians.

## 4 Dispute-key availability

Fix an epoch $E$, a record $r$ with KEM component $C_1$, and a request time $\tau$ on the bulletin board.

**Definition 1.** Ciphertext availability. $\mathrm{CA}(r, \tau)$ holds if the committed encrypted record and its receipt are retrievable from the DA layer at $\tau$.

**Definition 2.** Valid-share registration. $\mathrm{VSR}(E)$ holds if the registered public shares are consistent with one degree-$(t-1)$ polynomial whose constant term is the epoch secret, and every admitted custodian initially received the matching share.

**Definition 3.** Readiness. For a request $\mathsf{req}$, an audit or a dispute on a named record, $\mathsf{Ready}_i(E, \tau, \mathsf{req}, \Delta) = 1$ if custodian $i$ can access its current share, perform the requested operation, produce a valid DLEQ proof and deliver the response by $\tau + \Delta$. Audit readiness and dispute readiness for each record are distinct predicates.

**Definition 4.** Audit-time readiness. For a sample $S$ of $s$ custodians and threshold $q \le s$, $\mathrm{AKR}(E, S, \tau)$ holds if at least $q$ members of $S$ deliver valid canary responses by the audit deadline.

**Definition 5.** Authorized decryptability and deadline liveness. After a valid authorization for $r$ at $\tau$, $\mathrm{AD}(r, E, \tau)$ holds if the combiner obtains valid partial decryptions of $C_1$ from at least $t$ distinct custodians, and $\mathrm{DDL}(r, E, \tau, \Delta_d)$ holds if this happens by $\tau + \Delta_d$.

**Definition 6.** Pre-authorization confidentiality. PAC holds for $r$ if no probabilistic polynomial-time adversary that satisfies Assumption 1, chooses two equal-length plaintexts for $r$ and observes the sealed record and all audit transcripts, but no dispute releases, guesses which plaintext was sealed with non-negligible advantage. In the stronger $\mathrm{PAC}^+$, the adversary may also commit records of its choice and obtain the release of every record other than $r$, where a *release* is the set of partial decryptions and the combined value delivered to the authorized destination, which the adversary may control for its own records.

**Definition 7.** Dispute-key availability. Fix a readiness model, a distribution over custodian failures, share loss and message delays over time. An epoch provides $(\varepsilon, \Delta_d)$-DKA under it if for every committed record $r$ and authorization time $\tau$, $\Pr[\mathrm{DDL}(r, E, \tau, \Delta_d)] \ge 1 - \varepsilon$ over the readiness model and the protocol's randomness, and $\mathrm{PAC}^+$ holds for every record before its authorization.

**Definition 8.** Catastrophic state and detection. An epoch is *catastrophic* at $\tau$ if fewer than $t$ custodians are audit-ready. An audit *falsely accepts* if AKR holds in a catastrophic state; its *detection probability* is one minus its false-accept probability. Dispute readiness can be lower than audit readiness (Observation 1).

*Remark 1.* Timescale and epochs. Dispute windows last days (about 6.4 days on Arbitrum One[19]) and interactive moves hours, so DKA chiefly requires the key to stay serviceable throughout the window, while $\Delta_d$ matters for per-move timers. A record stays disputable after its committee rotates, so old committees must stay serviceable, or hand over shares by dynamic proactive sharing[20,21], and audits must cover every epoch with open records.

The next observation records what the weaker properties do not imply; each elementary witness blocks a tempting inference.

**Observation 1.** Non-implications. *Fix* $1 \le t \le n$, $1 \le q \le s \le n$ *and* $\Delta_d$.
*1.* $\mathrm{CA}(r, \tau)$ *does not imply* $\mathrm{DDL}(r, E, \tau, \Delta_d)$.

*2.* $\mathrm{AKR}(E, S, \tau)$ *does not imply* $\mathrm{AD}(r, E, \tau')$ *for an authorized dispute at* $\tau' \geq \tau$.
*3. For every* $m$, $m$ *accepted audits at times* $\tau_1 < \cdots < \tau_m$ *do not imply* $\mathrm{AD}(r, E, \tau')$ *for* $\tau' > \tau_m$.

**Proof.** (1) Let the DA layer retain the record while exactly $t-1$ custodians are dispute-ready for $r$ and the others have lost their shares. CA holds, but at most $t-1$ valid partials exist, so AD and DDL fail. (2) Let all custodians be audit-ready and let $n-t+1$ of them answer every canary but withhold every response for $r$: AKR holds for every $S$, while at most $t-1$ custodians answer the dispute. (3) Take two executions with all custodians audit-ready up to $\tau_m$; afterwards all stay ready in the first and $n-t+1$ lose their shares in the second. The audit histories are identical but AD holds at $\tau'$ only in the first, so without a model of how readiness evolves no predicate of the history implies AD.

Item (2) is why audit and dispute readiness are separate predicates; item (3) is why a repeated-audit guarantee needs a temporal model, static (Corollary 1) or stochastic (Section 6.2).

## 5 The KEYSTONE protocol

*Setup and sealing.* The committee is formed under identity, stake and failure-domain placement policy (Observation 2). A deployment would use an authenticated distributed key generation or publicly verifiable secret sharing[6,22,23]; the prototype uses a dealer with Feldman commitments as an explicit substitution boundary. To seal $r$ under the label $L_r = (E, \mathsf{id}_r, \text{policy hash})$, the worker draws $\rho$, computes $C_1 = g^\rho$ and $\kappa = \mathrm{KDF}(PK_E^\rho, L_r)$, wraps $K_r$ under $\kappa$, adds a TDH2 proof that binds $L_r$ and the wrapped key to $C_1$ [15], encrypts the record under $K_r$ with AES-GCM, and publishes the envelope with its DA pointer.

*Readiness audit.* The coordinator commits an epoch and audit slot. The next finalized beacon value $\beta$ fixes a sample $S$ of $s$ custodians, uniform or stratified by domain, and everyone derives the canary $X = H_G(\texttt{KEYSTONE-CANARY} \parallel E \parallel \mathsf{slot} \parallel \beta)$. Each sampled custodian returns $D_i = X^{x_i}$ with a DLEQ proof that $\log_g Y_i = \log_X D_i$, bound to the request context and its index. The verifier checks membership, freshness, uniqueness per index, proof validity and the deadline, and accepts after $q$ valid responses.

*Authorized dispute.* A dispute starts once the policy contract binds a record, epoch, authorized verifier and deadline $\Delta_d$. Each custodian checks the authorization, the TDH2 proof and that the label is the authorized record's, and returns $C_1^{x_i}$ with a DLEQ proof bound to the dispute. The combiner verifies responses on arrival, off its network event loop, discards duplicates and invalid ones, stops at $t$ valid partials, interpolates $C_1^x = PK_E^\rho$, re-derives $\kappa$, unwraps $K_r$ and decrypts.

*Transcript and evidence.* For each request, the board records commitments to valid, invalid, equivocal, stale and unauthorized responses and the absence of missing ones, with timestamps. Invalid proofs and conflicting responses are self-contained cryptographic evidence; attributing a missing response to a custodian additionally requires Assumption 4. The prototype contract is a skeleton: an administrator supplies the sample and canary hash when opening an audit without an on-chain beacon check, records invalid-response evidence after off-chain verification, and can cancel requests. Theorem 1 therefore holds for the prototype only if the administrator samples as the protocol prescribes; deriving the sample on chain from the committed beacon and verifying proofs on chain are needed for a trustless deployment.

## *5.1 Security of the audit*

**Lemma 1.** Correctness and knowledge soundness. *An honest response passes verification. In the random-oracle model a verifying response satisfies $D_i = X^{x_i}$ except with negligible probability, and the proof is a proof of knowledge: from any party that produces verifying responses for fresh canaries with non-negligible probability, an extractor obtains $x_i$. Any $t$ verified partials of $C_1$ with distinct indices $I$ combine to $\prod_{i\in I}(C_1^{x_i})^{\lambda_i^I} = PK_E^{\rho}$.*

**Proof sketch.** Completeness and combination are direct computations. The protocol is special-sound, and with the Fiat–Shamir hash as a random oracle the forking lemma yields an extractor[16,17]. Full proofs of Lemmas 1–3 are in the supplementary material.

Knowledge soundness makes a passed audit meaningful: whoever passes canary audits with non-negligible probability holds $x_i$, and under Assumption 1 the adversary cannot obtain an honest custodian's share from public values without solving a discrete logarithm.

**Lemma 2.** Audit transcripts are simulatable. *Model $H_G$ and $H$ as random oracles. A simulator that knows only $g$ and $Y_1, \ldots, Y_n$ answers all queries to $H_G$ and $H$ and produces the honest custodians' audit responses $(X, (D_i, \pi_i)_{i\in S})$ for any audit context and any $S$, including $S = \{1, \ldots, n\}$. With at most $N$ simulated proofs and $Q$ oracle queries, simulated and real views are within statistical distance $NQ/\ell$, so observing audits changes any adversary's advantage against* PAC *by at most $2NQ/\ell$.*

**Proof sketch.** The simulator answers each fresh $H_G$ query with $g^a$ for a stored random $a$, so for an honest custodian and canary $X = g^a$ it can output the real response $D_i = Y_i^a = X^{x_i}$, and it produces each proof with the honest-verifier zero-knowledge simulator, programming $H$; a programmed input was queried before with probability at most $Q/\ell$. Domain separation keeps this programming apart from a KEM reduction that programs the key-derivation oracle.

This is the standard simulation argument for threshold-coin and threshold-signature shares on hashed messages[10,11], which we restate for the audit setting. Lemma 2 holds even for adversarially chosen audit contexts and beacons. It is why the audit can exercise every sampled share without opening any record: combining $t$ canary partials yields $X^x$ for a hash-derived $X$, which equals a fixed record component only with negligible probability.

**Lemma 3.** No production substitution. *Model $H_G$ as a random oracle. For a fixed $C_1$, a coordinator making $Q$ queries to $H_G$ finds an audit context whose canary equals $C_1$ with probability at most $Q/(\ell - 1)$, by a union bound over uniform oracle answers.*

*Remark 2.* The prototype hashes to ristretto255 with the standard map from a 64-byte hash[12]; the ffdhe baseline uses `expand_message_xmd`[24] before squaring. Binding request context into the Fiat–Shamir hash does not give the share isolation of context-dependent threshold decryption[25], and the lemmas do not cover side channels.

### 5.2 Confidentiality against releases of other records

Lemma 2 concerns audits only. An adversary that can also obtain releases, for example a worker that commits and disputes its own records, defeats the plain KEM.

**Proposition 1.** Malleability across records. *With the plain threshold Diffie–Hellman KEM,* $\mathrm{PAC}^{+}$ *does not hold: an adversary that obtains the release of one record of its choice recovers* $K_r$ *for any target record* $r$*.*

**Proof.** For the target's component $C_1$, the adversary commits a record $r'$ with $C_1' = C_1 g^{\gamma}$ and obtains its release $C_1'^{x} = C_1^{x} PK_E^{\gamma}$. It computes $PK_E^{\rho} = C_1^{x} = C_1'^{x} PK_E^{-\gamma}$, derives the target's key-encryption key from $PK_E^{\rho}$ and the target's public record identifier and metadata, and unwraps $K_r$. Already $\gamma = 0$, re-submitting $C_1$ under a new identifier, suffices.

The attack does not involve the audit. It is blocked by a threshold KEM whose custodians refuse components not bound to the authorized record, such as TDH2 with the record label[15] or context-dependent threshold decryption[25]; KEYSTONE uses TDH2.

The dispute ciphertext is a tag-KEM/DEM hybrid[26]: TDH2's threshold key encapsulation, whose proof hashes the label and the wrapped key as a tag, followed by AES-GCM under $\kappa = \mathrm{KDF}(PK_E^{\rho}, L)$.

**Assumption 5.** TDH2 tag-KEM. This tag-KEM is secure against chosen-ciphertext attacks in the threshold model of Shoup and Gennaro, who prove the corresponding property of TDH2 encryption under DDH in the random-oracle model[15]; threshold tag-KEMs with this property are also constructed in that model[27].

**Lemma 4.** Record-bound releases. *Let the dispute KEM be TDH2 with label* $L_r$*, and let each custodian release* $C_1^{x_i}$ *only for a ciphertext whose TDH2 proof verifies and whose label is the authorized one. Under Assumptions 1 and 5 and static corruption,* $\mathrm{PAC}^{+}$ *holds for every record, and Lemma 2 continues to hold.*

**Proof sketch.** A release of a record $r' \neq r$ answers a decryption-share query for a valid ciphertext with label $L_{r'} \neq L_r$. Because the proof hashes the tag, a re-labelled or mauled $C_1$ either fails verification and is refused, or is a valid ciphertext other than the challenge, which the chosen-ciphertext game of Assumption 5 answers. The tag-KEM/DEM composition theorem[26] then gives chosen-ciphertext security of the hybrid with AES-GCM, so the target's wrapped key and record stay hidden. The canary path is unchanged and its oracles are domain separated, so the simulation of Lemma 2 composes with the TDH2 reduction.

The cost is one ciphertext-proof check per custodian per dispute (Section 7.4). The prototype's custodians enforce both checks, and a test reproduces the attack of Proposition 1 once they are removed.

## 6 Analysis

**Theorem 1.** False-accept probability. *Suppose exactly* $k$ *custodians are audit-ready throughout one audit, the sample* $S$ *of size* $s$ *is uniform without replacement and independent of readiness*

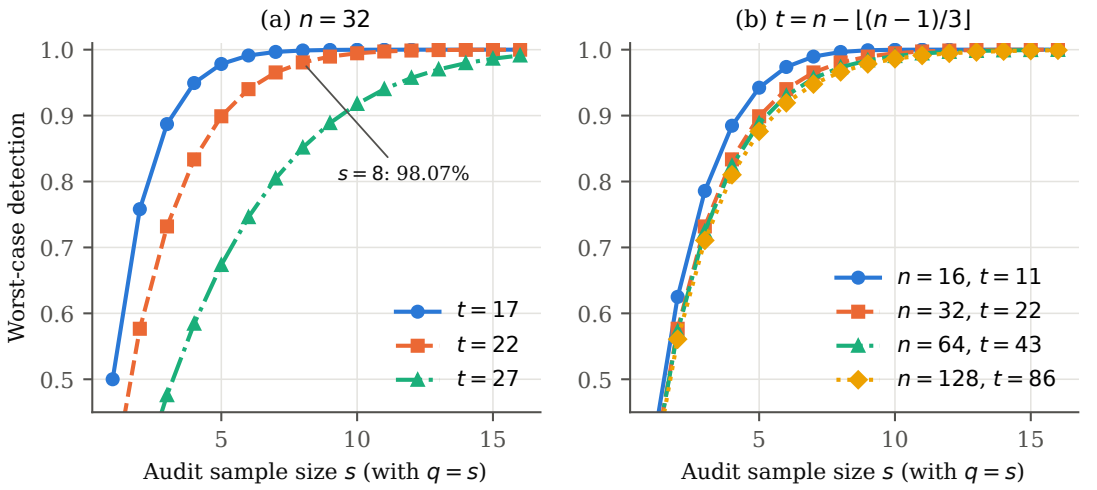



**Figure 2.** Exact worst-case detection $1 - \binom{t-1}{s}/\binom{n}{s}$ versus sample size with $q = s$. (a) Thresholds $t \in \{17, 22, 27\}$ at $n = 32$; the baseline $s = 8$, $t = 22$ is annotated. (b) Committee sizes 16–128 with $t = n - \lfloor (n-1)/3 \rfloor$ (values for four sizes in supplementary Table S1).

*(Assumption 2), and acceptance requires $q$ valid responses. Then, up to the negligible soundness error of Lemma 1,*

$$\Pr[\text{accept} \mid k] = \sum_{j=q}^{\min(s,k)} \frac{\binom{k}{j}\binom{n-k}{s-j}}{\binom{n}{s}}. \tag{1}$$

*This is non-decreasing in $k$, so over catastrophic states ($k \le t-1$) it is maximized at $k = t-1$, and for $q = s$*

$$\varepsilon_a = \max_{k<t} \Pr[\text{accept} \mid k] = \frac{\binom{t-1}{s}}{\binom{n}{s}}. \tag{2}$$

**Proof.** The number of ready custodians in $S$ is hypergeometric $(n, k, s)$, and by Lemma 1 the audit accepts exactly when it is at least $q$, up to negligible error. If $R \subset R'$, then $|S \cap R| \le |S \cap R'|$ for every $S$, which gives monotonicity; for $q = s$ only $j = s$ contributes.

For $n = 32$, $t = 22$, $s = q = 8$, $\varepsilon_a = \binom{21}{8}/\binom{32}{8} = 203{,}490/10{,}518{,}300 = 0.0193$, so a single audit detects the worst catastrophic state with probability 0.9807 under a static, non-selective fault model. Section 6.1 shows why this number alone is misleading. Figure 2 shows detection against $s$.

**Corollary 1.** Repeated audits under a static state. *If the ready set is fixed and catastrophic for $m$ audits whose samples are independent (Assumption 2), all $m$ falsely accept with probability at most $\varepsilon_a^m$.*

**Table 1.** Exact acceptance-rule trade-off at $n = 32$, $t = 22$: worst-case detection ($k = t - 1$, Theorem 1) versus the probability that a healthy committee passes under IID unavailability $f$ (Observation 3). Post hoc analysis.

| | | Worst-case | Benign audit pass under IID unavailability | | |
|---|---|---|---|---|---|
| $s$ | $q$ | detection | $f = 0.05$ | $f = 0.10$ | $f = 0.20$ |
| 8 | 6 | 57.5% | 99.4% | 96.2% | 79.7% |
| 8 | 7 | 85.9% | 94.3% | 81.3% | 50.3% |
| 8 | 8 | 98.1% | 66.3% | 43.0% | 16.8% |
| 12 | 10 | 89.6% | 98.0% | 88.9% | 55.8% |
| 12 | 11 | 98.2% | 88.2% | 65.9% | 27.5% |
| 12 | 12 | 99.9% | 54.0% | 28.2% | 6.9% |
| 16 | 13 | 93.2% | 99.3% | 93.2% | 59.8% |
| 16 | 14 | 98.8% | 95.7% | 78.9% | 35.2% |
| 16 | 15 | 99.9% | 81.1% | 51.5% | 14.1% |

**Proof.** Given the fixed ready set, the acceptance events are independent, each with probability at most $\varepsilon_a$ by Theorem 1.

By Observation 1(3), the static hypothesis cannot be dropped without a temporal model.

For samples stratified by domain, the per-domain counts of ready sampled custodians are independent hypergeometric variables, so acceptance is the probability that their sum reaches $q$ (supplementary Proposition S1); the worst catastrophic state then depends on how the $t - 1$ ready custodians are spread across domains.

**Observation 2.** Placement cap. *With no failures other than whole-domain outages, no single domain outage prevents reconstruction if and only if every domain has at most $n - t$ custodians, since an outage of a domain of size $d$ leaves $n - d$.*

The baseline places four domains of eight under the cap $n - t = 10$: one outage is survivable, two leave $16 < 22$ custodians.

**Observation 3.** Benign pass probability. *If each custodian is independently unavailable with probability $f$ and the sample is independent of availability, the audit passes with probability* $\Pr[\mathrm{Bin}(s, 1 - f) \geq q]$*, since the sampled custodians' availability indicators are independent.*

The strict rule $q = s$ that maximizes detection also turns every benign miss into an alarm. For $s = q = 8$ and $f = 0.10$ a healthy committee passes only with probability $0.9^8 = 0.430$, although $t$ of 32 shares are reachable with probability $\Pr[\mathrm{Bin}(32, 0.9) \geq 22] = 0.99983$. Table 1 and supplementary Figure S1 give the trade-off. At $s = 8$, $q = 7$ lowers detection to 85.9% and raises the benign pass rate at $f = 0.10$ only to 81.3%; a larger sample helps: $s = 16$, $q = 14$ detects 98.8% with benign pass rates of 95.7% at $f = 0.05$ and 78.9% at $f = 0.10$. This post hoc analysis is design guidance, not a prespecified result.

**Table 2.** Exact operating characteristic: probability that an audit with sample size $s$ and threshold $q$ accepts when $k$ of $n = 32$ custodians are ready ($t = 22$, uniform sampling, Theorem 1).

| | | Pr[audit accepts] with $k$ ready custodians ($n = 32$, $t = 22$) | | | | | |
|---|---|---|---|---|---|---|---|
| $s$ | $q$ | $k = 21$ | $k = 22$ | $k = 24$ | $k = 26$ | $k = 28$ | $k = 30$ |
| 8 | 8 | 1.9% | 3.0% | 7.0% | 14.9% | 29.5% | 55.6% |
| 8 | 7 | 14.1% | 19.3% | 33.3% | 52.4% | 74.6% | 94.4% |
| 16 | 14 | 1.2% | 2.7% | 11.0% | 32.7% | 70.0% | 100.0% |
| 16 | 13 | 6.8% | 12.6% | 34.3% | 67.3% | 94.9% | 100.0% |
| 24 | 20 | 0.1% | 0.5% | 8.2% | 47.6% | 100.0% | 100.0% |

## 6.1 Operating characteristic and sequential audits

Worst-case detection hides how weak a small audit is near the threshold. Table 2 gives the exact operating characteristic: with $s = q = 8$ the audit accepts a committee one share short of $t$ with probability 1.9%, a live committee with no spare share with 3.0%, and one with four spare shares with only 14.9%. Only the full probe ($s = n$) separates $k = t - 1$ from $k = t$ at 5% error rates, so sampled guarantees must be stated for a safety margin $m$, testing $k \le t - 1$ against $k \ge t + m$. With error rates of 5% each, a single audit needs $s = 25$ ($q = 19$) for $m = 4$, $s = 20$ ($q = 16$) for $m = 6$ and $s = 14$ ($q = 12$) for $m = 8$. A sequential probability ratio test across audits[28], valid under the static state of Corollary 1, is far cheaper: for $m = 6$ it decides on average after 2.3 audits for a failed committee and 1.1 for a live one with $s = 16$, $q = 14$, and after 9.8 and 4.6 with $s = q = 8$ (Wald's approximations). Audits should therefore be specified by the margin they certify and their error rates, not by worst-case detection.

**Proposition 2.** Deadline budget. *Under Assumptions 3 and 4, suppose at least $t$ custodians are non-faulty, hold their shares and follow the protocol, and the combiner is work-conserving, verifies serially and verifies each authenticated index at most once. Then a dispute requested at $\tau$ obtains $t$ valid partials by $\tau + \delta_b + c_p + \delta + n\,c_v$, and the key is recovered $c_c$ later. Hence $\Delta_d \ge \delta_b + c_p + \delta + n\,c_v$ suffices for* DDL*, and $n\,c_v$ may be replaced by $t\,c_v$ if no invalid response precedes the $t$-th valid one.*

**Proof.** Each of the $t$ correct custodians' responses arrives by $T' = \tau + \delta_b + c_p + \delta$. Authenticated channels stop an adversary from occupying a correct custodian's index, so the work-conserving combiner verifies at most $n$ responses and finishes by $T' + n\,c_v$, or by $T' + t\,c_v$ if no invalid response comes first.

On a production settlement layer, $\delta_b$ is a block or finality time and can dominate. With random delays, a per-message quantile for $\delta$ does not budget for the slowest of the $t$ responses needed:

**Corollary 2.** Stochastic deadline. *Suppose $n_\ell$ custodians can answer, each response is delivered independently with probability $p_\ell$ after a round trip with distribution function $F$, and the combiner finishes within $B$ after the $t$-th valid response. Then* $\Pr[\mathrm{DDL}(r, E, \tau, \Delta_d)] \ge \Pr[\mathrm{Bin}(n_\ell, p_\ell F(\Delta_d - B)) \ge t]$.

**Proof sketch.** The number of responses delivered within $\Delta_d - B$ is $\mathrm{Bin}(n_\ell, p_\ell F(\Delta_d - B))$; if it reaches $t$, the dispute completes by $\Delta_d$.

### 6.2 Readiness over time

We now let custodians fail and recover. Each custodian is an independent two-state Markov chain with failure rate $\phi$ and repair rate $\psi$, so it is up a fraction $v = \psi/(\phi + \psi)$ of the time; in a variant, each of four domains of eight also fails as a unit at rate $\phi_d$ and recovers at rate $\psi_d$. Audits run every $T$ with a fresh uniform sample and only observe: they trigger no repair. Faults are non-selective, so a custodian is audit-ready exactly when it is dispute-ready, and disputes arrive independently of the process. Let $Z(\tau)$ be the number of ready custodians, $A(k)$ the acceptance probability of Eq. (1), $\omega$ the stationary law of $Z$, and $\mathbf{1}_{<t}$ the indicator column of $Z < t$; vectors are rows over $k \in \{0, \dots, n\}$.

**Theorem 2.** Risk given an audit history. *$Z$ is a Markov chain with $P_u(k, k') = \sum_i \mathrm{b}(i; k, p_\uparrow(u))\, \mathrm{b}(k' - i; n - k, p_\downarrow(u))$, where* b *is the binomial mass function, $p_\uparrow(u) = v + (1 - v)e^{-(\phi+\psi)u}$ and $p_\downarrow(u) = v(1 - e^{-(\phi+\psi)u})$. For audit outcomes $o_1, \dots, o_j$, let $\alpha_1 \propto \omega \odot e_{o_1}$ and $\alpha_i \propto (\alpha_{i-1} P_T) \odot e_{o_i}$, normalized, with $e_{\mathsf{acc}} = A$ and $e_{\mathsf{rej}} = 1 - A$. A dispute at $jT + u$, $0 \le u < T$, fails to reach $t$ with probability $\alpha_j P_u \mathbf{1}_{<t}$ given $o_1, \dots, o_j$. With domain shocks, the same holds for the chain of per-domain pairs (domain up, members up).*

**Proof sketch.** From any configuration with $k$ custodians up, after time $u$ each up custodian is up with probability $p_\uparrow(u)$ and each down one with $p_\downarrow(u)$, independently, so $Z$ is strongly lumpable with kernel $P_u$. A fresh uniform sample makes each outcome conditionally independent of everything else given $Z$ at its audit, with emission $A$, so the history is a hidden Markov model and the recursion is the forward algorithm.

After an unbroken run of accepted audits, $\alpha_j$ converges from any start to the normalized Perron vector $\alpha^*$ of $P_T \operatorname{diag}(A)$ on the states the audit can accept, where it is entrywise positive for $T > 0$. We call $R(T) = \sup_{0 \le u < T} \alpha^* P_u \mathbf{1}_{<t}$ the *posterior risk*: the failure probability a dispute faces, at the worst point of an audit cycle, after a long clean audit record.

**Theorem 3.** Undetected failure. *Let $U(T) = \frac{1}{T}\int_0^T (\omega \odot A) P_u \mathbf{1}_{<t}\, du$, the long-run probability that a dispute fails and the most recent audit accepted. With or without domain shocks, $U$ is non-decreasing in $T$ and $U(T) \ge U_0 = \sum_{k<t} \omega(k) A(k)$, its limit as $T \to 0$. Without domain shocks,*

$$U(T) \le A(t-1)\, \omega(Z < t) + \sum_{k \ge t} \omega(k) A(k) \Pr[\mathrm{Bin}(k, p_\uparrow(T)) < t].$$

**Proof sketch.** The chain is reversible, moves one custodian or domain at a time, and is stochastically monotone because $p_\uparrow \ge p_\downarrow$. Let $g(u) = (\omega \odot A) P_u \mathbf{1}_{<t}$. Then $g'(u)$ is minus the Dirichlet form of $A$ and $P_u \mathbf{1}_{<t}$; $A$ is increasing and $P_u$ keeps $\mathbf{1}_{<t}$ decreasing, so every term has the sign that makes $g' \ge 0$. Hence $U$, the mean of $g$ over $[0, T]$, is non-decreasing and at least $g(0) = U_0$. For the bound, use $A(k) \le A(t-1)$ for $k < t$, and for $k \ge t$ note that the custodians up at the audit and still up at $u$ number $\mathrm{Bin}(k, p_\uparrow(u))$, lower-bound $Z(u)$, and decrease in $u$.

Since audits trigger no repair, the failure probability itself is $\omega(Z < t)$; $U$ measures how much of it the latest audit fails to flag. A target $\varepsilon < U_0$ cannot be met by auditing more often, only by a stronger design or a larger availability margin. Otherwise the longest admissible interval

$T^*(\varepsilon) = \sup\{T : U(T) \le \varepsilon\}$ is the cheapest schedule, found by bisection because $U$ is monotone, and the bound gives a conservative interval in closed form. Full proofs are in the supplementary material.

*Repair on alarm.* So far audits only observe. Let a rejected audit trigger a resharing after a delay $D_r \le T$. Resharing needs at least $t$ ready custodians, because fewer shares cannot reconstruct the epoch secret; if it succeeds, every custodian is ready again, and if not, nothing happens until the next audit. A domain outage persists, since resharing within the committee does not end it.

**Proposition 3.** Repair on alarm. *The state just before audits is a Markov chain with kernel $M = \mathrm{diag}(A)P_T + \mathrm{diag}(1-A)P_{D_r}\Pi_t P_{T-D_r}$, where $\Pi_t$ makes every custodian ready in states with $Z \ge t$ and leaves the others unchanged. With $\mu$ its stationary law, $\mu_a = \mu \odot A$ and $\mu_r = \mu \odot (1-A)$, the long-run probability that a dispute fails is*

$$F_{\mathrm{rep}}(T) = \frac{1}{T}\Big[\int_0^T \mu_a P_u \mathbf{1}_{<t}\,du + \int_0^{D_r} \mu_r P_u \mathbf{1}_{<t}\,du + \int_0^{T-D_r} \mu_r P_{D_r}\Pi_t P_u \mathbf{1}_{<t}\,du\Big],$$

*alarms occur at rate $\mu_r \mathbf{1}/T$ and resharings at rate $\mu_r P_{D_r}\mathbf{1}_{\ge t}/T$.*

**Proof sketch.** Between audits the state evolves by $P_u$; at an audit it branches on the outcome, and a rejection inserts the reset after $D_r$. The pre-audit states therefore form a Markov chain with the primitive kernel $M$, and the renewal-reward theorem averages failure over an interval started from $\mu$; the three integrals cover an accepted interval, the repair window and the rest of a rejected interval, after a resharing if one was possible.

Unlike $U$, $F_{\mathrm{rep}}$ falls as audits become more frequent, but every rejection costs a resharing, benign ones included; Section 7.5 quantifies this trade-off.

## 6.3 Selective withholding

Observation 1(2) shows that withholders can pass every audit and defeat a dispute. The next proposition bounds what any test can do about it.

**Proposition 4.** Targeted withholding. *A custodian that answers every request except those for one record $r$ it can identify produces, on every audit, decoy or other request not for $r$, exactly the transcript of an honest custodian, so no such test detects it. If a fraction $\theta$ of dispute requests are decoys over protocol-sealed records that custodians cannot tell apart from real ones, a custodian that refuses a class of disputes it cannot separate from decoys is caught, under Assumption 4, with probability $\theta$ for each refusal.*

**Proof sketch.** On requests other than $r$ the custodian behaves honestly, so its transcript is the honest one. In the second case each refused request is a decoy with probability $\theta$ independently of the custodian's view, and the party that sealed the decoy observes the missing response by the deadline.

Decoys are not free: a decoy fraction $\theta$ multiplies the dispute load on custodians by $1/(1-\theta)$.

A known target is therefore protected only by deterrence after the fact: the deadline transcript names the non-responders under Assumption 4. If blocking needs $w$ withholders and each forfeits a stake $S$ with probability $p_a$, blocking a dispute worth $V$ is unprofitable when $V < w\,S\,p_a$; Section 7.6 gives the smallest $w$.

# 7 Evaluation

We compare the audit with simpler checks (Q1), separate audit outcomes from dispute success under failures (Q2), test deadline liveness (Q3), measure cost (Q4), evaluate readiness over time with and without repair (Q5) and quantify selective withholding (Q6).

*Method.* Table 3 lists every experiment, and each result carries one of four evidence labels. *Exact* values are closed forms or recursions under a declared model. *Simulation* results come from cells whose parameters, endpoints and seeds were fixed and hashed before execution; the three Q2 cells have 131,072 draws each, above the 73,778 that Hoeffding's inequality requires for a proportion's half-width of 0.005 at 95% confidence[29], and where an exact value exists simulation serves as an implementation check. *Microbenchmarks* are single-process medians, and *emulation* runs 32 custodian processes on one host. Each run was specified in a design hashed and publicly timestamped before it ran (verifiable after review). R1, the first prototype (ffdhe2048, serial verification on the network loop), produced Q1, Q2 and the comparison column of Q3; R2 (ristretto255, TDH2, verification off the loop) produced Q3 to Q5; R2b replicated Q3 on fresh traces with a corrected deadline; and R2c and its correction R2d produced Q6 and the repair results of Q5. Analyses computed after a run are labeled post hoc. Proportions carry 95% Wilson intervals[30].

## 7.1 Readiness baselines (Q1)

A heartbeat is answered even by a custodian that lost its share, registration is checked once, and decrypting a real record discloses it (supplementary Table S4 compares these checks). A full-committee canary probe ($s = n$) detects every static catastrophic state at four times KEYSTONE's cost and is affordable once per epoch at $n = 32$. Sampling pays off when audits are frequent or recorded on chain: at 57,882 gas per response (supplementary Table S2), an audit costs about 0.46 million gas for $s = 8$ and 1.85 million for $s = 32$.

## 7.2 Robustness under failures (Q2)

In the simulation cells (Table 4), 10% independent unavailability leaves the committee reconstructable in 99.98% of draws while the strict audit passes in only 43.1%: a sensitive alarm, not a health score. Under 20% domain outages, stratified sampling raised conditional detection of catastrophic draws from 99.89% to 100.00% and lowered the benign pass rate from 37.3% to 34.9%. The prespecified endpoint, the paired difference in detected catastrophes per draw, was 0.00021 with Hoeffding half-width 0.0075, too conservative to resolve an effect this small; a post hoc exact sign test on the 28 discordant pairs, all favoring stratification, gives $p = 7.5 \times 10^{-9}$. With eleven selective withholders, audits passed in 100.0% of draws and disputes succeeded in 0.0%, reproducing the witness of Observation 1(2).

**Table 3.** Experimental configuration. The baseline committee is $n = 32$, $t = 22$, $s = q = 8$ with four failure domains of eight custodians.

| Experiment family | Parameters | Execution and evidence label |
|---|---|---|
| Static bound (Eq. 2) | $n \in \{16, 32, 64, 128\}$, $t = n - \lfloor (n-1)/3 \rfloor$, $s \in \{1, \ldots, 16\}$, $q = s$; baseline $n = 32, t = 22, s = q = 8$ | Exact closed form |
| Simulation cells | $n = 32, t = 22, s = q = 8$, 4 domains of 8. IID: $p_{\text{off}} = 0.10$. STRAT: $p_{\text{dom}} = 0.20$, $p_{\text{off}} = 0.02$, uniform vs. 2-per-domain quota. SW: $w = 11$ | 32 seed blocks × 4,096 draws = 131,072 per cell; seeds frozen before execution; simulation |
| Correlation model | $f \in \{0, 0.025, \ldots, 0.40\}$; IID vs. 2/4/8 equal domains | Exact; declared synthetic model |
| Acceptance rule | $s \in \{8, 12, 16\}$, $q \in [\lceil s/2 \rceil, s]$, IID $f \in \{0.05, 0.10, 0.20\}$ | Exact; post hoc |
| Cryptographic cost | R1: ffdhe groups, $n \leq 128$. R2: ristretto255 and ffdhe, 1,000 runs | Microbenchmark, CPython 3.13, Apple M4 Pro |
| Deadline testbed | R2 and R2b (fresh paired seeds): ristretto255, TDH2, $n = 32, t = 22$; 8 profiles × 300 traces × 2 variants; lognormal delay (median 0/40/120 ms), loss, crashes, domain outage | 32 processes on one host; emulation |
| Temporal readiness | MTBF 7/30/90 d, MTTR 1/6/24 h, $T$ 1/6/24 h, $(s, q)$ $(8, 8)/(16, 14)/(24, 20)$; domain shocks | Exact, synthetic; 3-cell simulation check |
| Repair and withholding (R2c) | Same grid with resharing after $D_r \in \{0.25, 1\}$ h; $w \leq 11$ withholders, $v \in \{0.9, 0.99, 0.999\}$ | Exact, synthetic; 3-cell simulation check |

**Table 4.** Simulation cells, 131,072 draws each, 95% Wilson intervals. Reconstructable means at least $t$ custodians can serve the dispute.

| Scenario | Audit pass | Reconstructable |
|---|---|---|
| IID 10% unavailable | 43.1% [42.8, 43.3] | 99.98% [99.97, 99.99] |
| Domain outage 20%, uniform sample | 37.3% [37.0, 37.6] | 81.3% [81.1, 81.6] |
| Domain outage 20%, stratified sample | 34.9% [34.7, 35.2] | 81.3% [81.1, 81.6] |
| 11 selective withholders | 100.0% [99.997, 100.000] | 0.0% [0.000, 0.003] |

Placement matters more (Figure 3). At per-unit unavailability $f = 0.10$, an IID committee reconstructs with probability 99.98%, four domains of eight (under the cap of Observation 2) with 94.8%, and two domains of sixteen, which violate it, with only 81.0%. Field studies show that correlated failures dominate real unavailability[31].

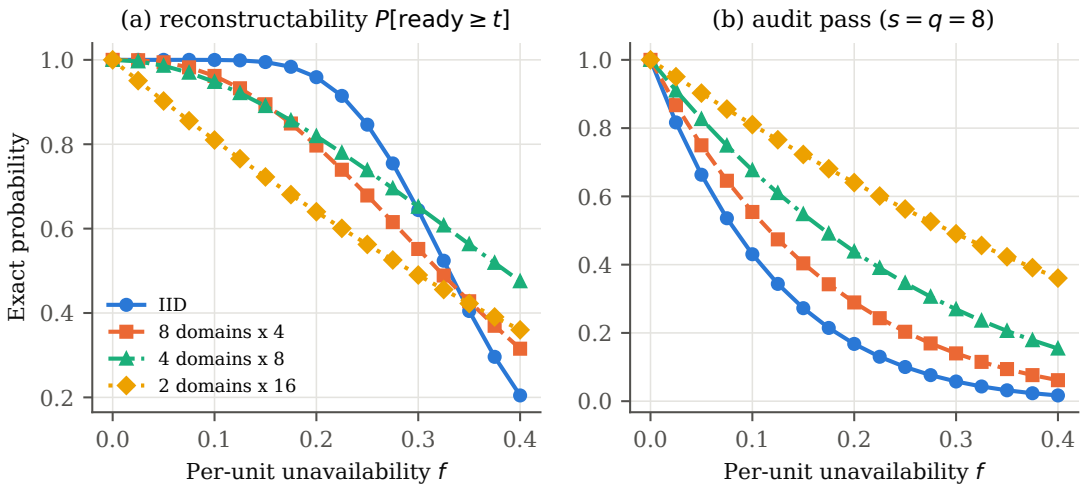



**Figure 3.** IID versus domain-correlated unavailability at $n = 32$, $t = 22$. A domain fails with probability $f$ and takes all its members; in the IID model each custodian fails independently with probability $f$. (a) Reconstructability. (b) Audit pass with $s = q = 8$. Exact; a 20,000-draw Monte Carlo check agrees within 0.009.

## 7.3 Deadline liveness in an emulated testbed (Q3)

Thirty-two custodian processes, each a TCP server holding only its own ristretto255 share, and one combiner run on one host. Each trace runs one canary audit ($s = q = 8$) and one dispute over a fresh 1 KiB record; the combiner verifies DLEQ proofs in a worker thread and stops at $t = 22$ valid partials. Delays are lognormal on both legs, lost messages are not retransmitted, and crashed custodians never answer. The testbed tests two things that do not depend on the delay scale: whether a deadline from Proposition 2 or Corollary 2 predicts completion, and where verification must run; dispute windows of days leave far more slack than its millisecond delays. Two runs of eight profiles × 300 traces × two verification variants used different deadlines (Table 5). R2 used $\Delta_d^* = c_p + 2\delta_{99} + t\,c_v + c_c$, Proposition 2 with each leg's 99th-percentile delay and the medians of Table 6, fixed before its run. R2b, designed after R2 and fixed before any of its traces, used fresh seeds shared by both variants and the deadline $\Delta_{os}$ at which Corollary 2 gives 0.99, treating the 99th percentile of a calibration run as $B$, so the 0.99 is approximate.

*R2.* Five of the seven reconstructable profiles succeeded in every trace, and eleven crashes ($21 < t$) defeated every dispute. With delay, completion followed the network, with medians from 106 to 601 ms, against 3.69–3.88 s in R1, where serial verification on the network loop took about 3 s. Two profiles missed. Without delay, no dispute met the 4.36 ms deadline (median 6.7 ms): the formula counts cryptographic work only, not message handling among 33 processes. With ten crashes all 22 survivors must answer, so the slowest of 22 round trips decides: 97.3%

**Table 5.** Emulated testbed: 32 custodian processes on one host, ristretto255 and TDH2, 300 traces per profile, variant and run. Success requires the correct plaintext by the run's deadline: $\Delta_d^*$ for R2, $\Delta_{os}$ (Corollary 2, calibrated $B$) for R2b, each fixed before its run; R2b serial with 95% Wilson interval; medians over R2b serial disputes that succeeded; R1 is the first prototype (ffdhe2048).

| | R2 (frozen deadline) | | | R2b (corrected deadline, fresh traces) | | | | R1 |
|---|---|---|---|---|---|---|---|---|
| Profile | $\Delta_d^*$ (ms) | serial | batch | $\Delta_{os}$ (ms) | serial | batch | median (ms) | median (ms) |
| control | 4.4 | 0.0% | 0.0% | 7.1 | 83.3% [78.7, 87.1] | 99.3% | 6.9 | 3818 |
| regional | 260 | 100.0% | 100.0% | 128 | 98.7% [96.6, 99.5] | 87.7% | 108 | 3841 |
| global | 974 | 100.0% | 100.0% | 408 | 98.7% [96.6, 99.5] | 98.3% | 325 | 3875 |
| global loss2 | 974 | 100.0% | 100.0% | 435 | 98.7% [96.6, 99.5] | 98.0% | 340 | 3829 |
| global crash10 | 974 | 97.3% | 94.7% | 1176 | 99.0% [97.1, 99.7] | 99.0% | 602 | 3692 |
| global crash11 | 974 | 0.0% | 0.0% | – | 0.0% [0.0, 1.3] | 0.0% | – | – |
| global domain outage | 974 | 100.0% | 100.0% | 663 | 99.3% [97.6, 99.8] | 99.0% | 464 | 3700 |
| heavy tail | 3918 | 100.0% | 100.0% | 785 | 100.0% [98.7, 100.0] | 100.0% | 464 | 3771 |

and 94.7% succeeded, and Corollary 2 with the frozen parameters, applied post hoc, predicts 96.3%.

*R2b.* A calibration run on the same harness gave the deterministic term $B = 7.09$ ms, the 99th percentile of disputes without delay (supplementary Section S6). In R2b, serial verification met the criterion, a Wilson upper bound of at least 0.99, in 7 of eight profiles, although $\Delta_{os}$ is far tighter than $\Delta_d^*$ (408 against 974 ms under global delay), and at the R2 deadline the ten-crash profile again succeeded in 96.3%, as Corollary 2 predicted. It missed without delay, 83.3% [78.7, 87.1], consistent with host drift between runs (median 6.83 in calibration, 6.9 ms in R2b), which we did not test: a deadline without network slack is fragile. In the paired design, batch verification was 8–12 ms slower at the median under delay and 0.9 ms faster without it (supplementary Section S6), contrary to the expected absence of a difference. Audit pass rates fell inside the Wilson interval of their exact expectation in 16 and 16 of 16 profile and variant pairs.

## 7.4 Cost and scalability (Q4)

Table 6 gives per-operation medians of the R2 prototype. In ristretto255 a partial decryption with proof takes 0.086 ms and its verification 0.15 ms, so the term $t\,c_v$ of Proposition 2 is 3.4 ms, against 1,062 ms for R2's own ffdhe2048 code and about 2,980 ms for R1's; combining 22 partials, unwrapping and decrypting takes 0.92 ms, and the TDH2 ciphertext-proof check of Lemma 4 adds 0.116 ms per custodian. Batch verification of 22 proofs by a random linear combination runs at 0.93 times the speed of serial verification in ristretto255, where scalar multiplications dominate either way, and at 1.83 and 2.37 times in ffdhe2048 and ffdhe3072. Supplementary Table S1 gives R1 costs for committees of up to 128.

## 7.5 Readiness over time (Q5)

We evaluated Theorems 2 and 3 on a synthetic grid of 243 cells (Table 3); Table 7 shows five scenarios. The bound of Theorem 3 held in all 81 shock-free cells. An event simulation without closed forms matched 29 of the 30 quantities with observed events within 99% Wilson intervals (3

**Table 6.** Median wall time per operation in the R2 prototype, 1,000 timed runs, CPython 3.13 on an Apple M4 Pro; ristretto255 through libsodium, RFC 7919 groups[13] in pure Python. Batch verification uses 128-bit random coefficients.

| Operation (median, ms) | ristretto255 | ffdhe2048 | ffdhe3072 |
|---|---|---|---|
| Canary derivation $H_G$ | 0.011 | 0.020 | 0.034 |
| Partial decryption + DLEQ proof | 0.086 | 49.53 | 160.8 |
| DLEQ verification | 0.152 | 48.26 | 140.0 |
| Batch verification of $t = 22$ proofs | 3.61 | 580.7 | 1301.6 |
| Seal 1 KiB, plain KEM | 0.041 | 32.38 | 103.9 |
| Seal 1 KiB, TDH2 | 0.106 | 83.29 | 266.5 |
| TDH2 ciphertext-proof check | 0.116 | 47.85 | 137.6 |
| Combine $t$ partials, unwrap, decrypt | 0.923 | 244.9 | 729.6 |

more, near $10^{-16}$, saw no events). Three findings stand out. First, correlated failures dominate: with custodian MTBF 30 d and MTTR 6 h, $\Pr[Z < t]$ is $1.4{\times}10^{-15}$ for independent custodians and $5.7{\times}10^{-5}$ when each domain fails every 90 days on average. Second, the floor binds: with domain shocks, custodian MTBF 7 d and MTTR 24 h, $U_0 = 1.1{\times}10^{-4}$ for $(8, 8)$, so no audit interval reaches $\varepsilon = 10^{-4}$, and $(24, 20)$ reaches it only with audits every 15 minutes, at the edge of the searched range. Third, larger samples are not uniformly better: $(24, 20)$ lowers the floor by orders of magnitude, yet its $U$ exceeds that of $(8, 8)$ in all 54 shock cells and 18 of 27 shock-free ones, each time because it accepts more often; $U$ ignores the benign alarms the strict rule raises instead. For independent custodians with MTBF 7 d, MTTR 24 h and $(8, 8)$, the longest interval for $\varepsilon = 10^{-4}$ is 22.8 h under $U$, 12.2 h under the posterior risk $R$, and a conservative 3.1 h from the bound.

*Repair on alarm.* With resharing after a rejected audit (Proposition 3, $D_r = 1$ h), the failure probability fell in all 486 grid rows, for example from $1.2{\times}10^{-3}$ to $1.6{\times}10^{-6}$ for independent custodians with MTBF 7 d, MTTR 24 h, $(8, 8)$ and $T = 6$ h, at 1.26 resharings a day (Figure 4). At least 90% of alarms came while at least $t$ custodians were ready, so a strict audit protects mainly by refreshing a committee before failures accumulate; because resharing needs $t$ ready custodians, it cannot rescue a committee that is already catastrophic. Table 8 turns this into a trade-off: for a target of $10^{-4}$, $(8, 8)$ needs the fewest audits and the most resharings and $(24, 20)$ the reverse; with domain shocks, only $(8, 8)$ with yearly domain failures reaches the target, and no design does with a 90-day domain MTBF, because resharing does not end a provider outage. The event simulation with repair matched 7 of the 7 quantities with observed events (2 saw none). Of 486 design rows, 36 were not monotone in $T$; the 3 with a bisected interval ($D_r = 0.25$ h, not tabulated) have a contiguous feasible range in a 200-point scan.

## 7.6 Selective withholding (Q6)

Figure 5 gives the exact effect of $w$ selective withholders when every other custodian is ready with probability $v$. With $v = 0.99$, ten withholders still leave the dispute 80.2% likely to succeed and eleven defeat it, while the strict audit passes more often than with none (94.9% against 92.3%), since withholders always answer canaries. Blocking needs 11 withholders at $v = 0.99$

**Table 7.** Readiness over time, exact under a declared synthetic model, $n = 32$, $t = 22$. Scenarios give custodian MTBF / MTTR and, for shocks, the domain MTBF (domain MTTR 6 h). $U$: probability that a dispute fails and the last audit accepted; $U_0$: its floor; $T^*$: longest interval with $U \le 10^{-4}$, by bisection over $[0.25\,\text{h}, 30\,\text{d}]$ (none: infeasible at 0.25 h).

| | | $U$, $T = 6$ h | | $U_0$ | | $T^*$ (h), $\varepsilon = 10^{-4}$ | |
|---|---|---|---|---|---|---|---|
| Scenario | $\Pr[Z < t]$ | (8,8) | (24,20) | (8,8) | (24,20) | (8,8) | (24,20) |
| IID, 7 d / 24 h | $1.2\times10^{-3}$ | $3.7\times10^{-5}$ | $2.4\times10^{-5}$ | $2.1\times10^{-5}$ | $6.2\times10^{-7}$ | 22.8 | 13.3 |
| IID, 30 d / 24 h | $2.7\times10^{-9}$ | $1.0\times10^{-10}$ | $8.1\times10^{-11}$ | $5.1\times10^{-11}$ | $1.7\times10^{-12}$ | $\geq$720 | $\geq$720 |
| Shock 365 d, 30 d / 6 h | $5.5\times10^{-6}$ | $1.2\times10^{-6}$ | $1.6\times10^{-6}$ | $5.5\times10^{-8}$ | $1.8\times10^{-9}$ | $\geq$720 | $\geq$720 |
| Shock 90 d, 30 d / 6 h | $5.7\times10^{-5}$ | $1.1\times10^{-5}$ | $1.3\times10^{-5}$ | $2.6\times10^{-7}$ | $7.1\times10^{-9}$ | $\geq$720 | $\geq$720 |
| Shock 90 d, 7 d / 24 h | $7.7\times10^{-3}$ | $6.9\times10^{-4}$ | $1.8\times10^{-3}$ | $1.1\times10^{-4}$ | $2.5\times10^{-6}$ | none | 0.3 |

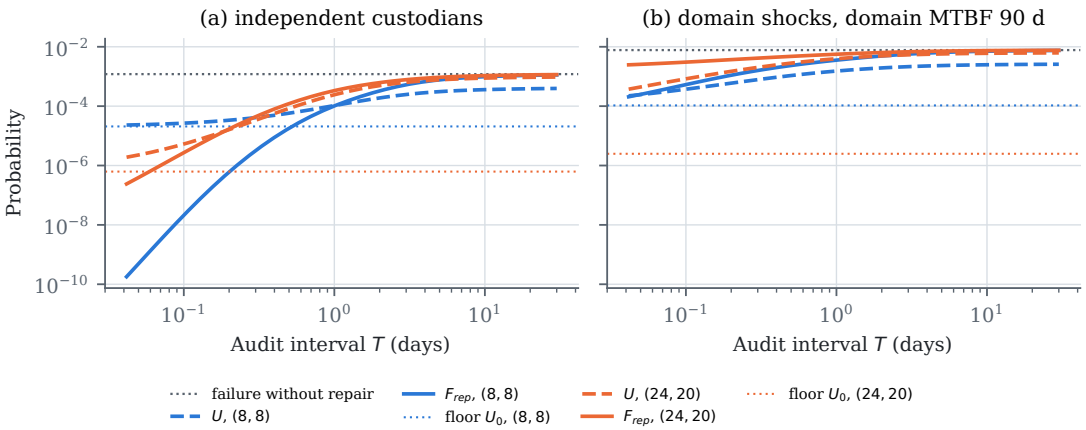



**Figure 4.** Readiness over time for custodian MTBF 7 d and MTTR 24 h, exact under the declared synthetic model ($n = 32$, $t = 22$). Dashed: $U(T)$, failure undetected by the latest audit, without repair; solid: $F_{\text{rep}}(T)$, failure with resharing after each rejected audit when at least $t$ custodians are ready ($D_r = 1$ h, Proposition 3); dotted: floors $U_0$ and, in grey, the failure probability without repair.

but only 9 at $v = 0.9$, because a coalition can wait for natural outages. Withholding by the minimal coalition is unprofitable if each withholder's stake is at least 0.09 to 0.18 times the value of the blocked dispute, for attribution probabilities from 1 to 0.5.

## 8 Discussion

Five design consequences follow. First, specify audits by the safety margin they certify, with error rates at $k \le t - 1$ and $k \ge t + m$ (Section 6.1). Second, placement under the cap $n - t$ matters more than the sampling policy, and correlated domain failures dominate the risk over time (Q5). Third, schedule audits from a risk target and couple them to repair: without action an audit only reports risk, and below the floor $U_0$ only a stronger design helps; with resharing on alarm, the choice of $(s, q)$ trades audits against resharings, most of them benign (Table 8).

**Table 8.** Repair on alarm, exact under the declared synthetic model: longest audit interval $T^*$ with failure probability at most $10^{-4}$, custodian MTBF 7 d, MTTR 24 h, $D_r = 1$ h, and the audits, alarms and resharings per day it costs (a resharing needs at least $t$ ready custodians); benign is the share of alarms raised while at least $t$ custodians were ready; none: unattainable at $T \geq 0.25$ h. Shock rows add domain outages with the stated domain MTBF.

| Scenario | $(s, q)$ | no repair | $T^*$ (h) | audits/day | alarms/day | resharings/day | benign |
|---|---|---|---|---|---|---|---|
| IID | (8,8) | $1.2\times10^{-3}$ | 23.8 | 1.01 | 0.55 | 0.55 | 99.97% |
| | (16,14) | | 14.2 | 1.69 | 0.32 | 0.32 | 99.93% |
| | (24,20) | | 10.8 | 2.23 | 0.19 | 0.19 | 99.85% |
| Shock 365 d | (8,8) | $2.8\times10^{-3}$ | 2.0 | 11.92 | 2.03 | 2.03 | 99.94% |
| | (16,14) | | none | – | – | – | – |
| | (24,20) | | none | – | – | – | – |
| Shock 90 d | (8,8) | $7.7\times10^{-3}$ | none | – | – | – | – |
| | (16,14) | | none | – | – | – | – |
| | (24,20) | | none | – | – | – | – |

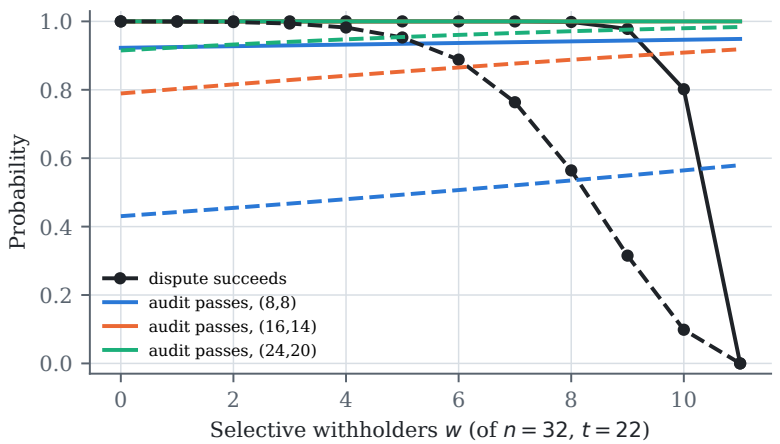



**Figure 5.** Selective withholding, exact: dispute success and audit pass against the number $w$ of withholders ($n = 32$, $t = 22$), with every other custodian ready with probability $v = 0.99$ (solid) or $v = 0.9$ (dashed). At $v = 0.99$ the $(16, 14)$ and $(24, 20)$ audits pass with probability near 1 for every $w$.

Fourth, deadlines must budget for the ledger delay, host overhead and the slowest of the $t$ needed responses (Corollary 2), not a per-message quantile; with elliptic-curve verification off the network loop, the network governs completion, and batch verification helps only when responses arrive together (Q3). Fifth, if untrusted parties can obtain releases, the dispute path must bind each release to its record (Proposition 1, Lemma 4).

Invalid proofs and equivocation are transferable evidence, whereas attributing non-response depends on timing[32]. The audit cannot detect selective withholding, because canaries are unrelated to records by design (Lemma 2), and making them look like record components would

stop observers from checking that an audit opened no record (Lemma 3); a known target is protected only by deterrence: decoy disputes catch withholding as a policy (Proposition 4), and stakes of roughly a tenth to a fifth of the disputed value per custodian make blocking unprofitable (Q6).

## 9 Related work

*Optimistic and verifiable computation.* Optimistic rollups rely on an honest challenger to force correct results on chain [1]; opML applies this to machine-learning inference [2], and EigenAI re-executes private inference in a trusted environment after a threshold key release [3]. These systems assume the challenger can read the disputed inputs; with encrypted inputs this becomes the requirement that keys be serviceable before the deadline, which KEYSTONE audits without changing the dispute game.

*Threshold decryption and key management.* Threshold cryptosystems [4], their chosen-ciphertext-secure variants [15] and DLEQ proofs [16,17] provide the release mechanism. Verifiable and publicly verifiable secret sharing and distributed key generation [5,6,22,23] make shares consistent when created, and proactive and dynamic sharing refreshes them [7,20,21]; a refresh also exercises every share, but it is infrequent and costly. A canary response is structurally a threshold-coin share [10] or a threshold-signature share on a hashed message [11]; our contribution is its use as a sampled readiness test and what that test can and cannot certify. Context-dependent threshold decryption isolates shares across contexts [25].

*Threshold release on ledgers.* CALYPSO releases on-ledger secrets to authorized readers through a threshold committee with an auditable access log [33], and ETHTID provides threshold disclosure on Ethereum [34]. Encrypted mempools face the same concerns of committees, epochs, collusion and timely decryption [35,36]. FairBlock has a committee exchange a fixed number of identity-based decryption shares per block [37], and F3B, building on CALYPSO, has a committee release a per-transaction key once the transaction is final [38]. In both, timely decryption depends on a threshold of members serving the release, the property DKA makes explicit; the descriptions we found specify the release path but no pre-release readiness audit. These ideas are established; KEYSTONE adds the deadline-bound property that release must satisfy for disputes, an audit that measures it without opening records, and an analysis of what passing it implies.

*Proofs of storage, custody and heartbeats.* Proofs of retrievability and provable data possession spot-check stored files and bound the probability that a server missing data passes [8,39], also under a mobile adversary [40]. KEYSTONE's audit belongs to this family, with a proof of knowledge (Lemma 1) and sampling bounds (Theorem 1). It differs in that possession is not service (Observation 1), in that confidentiality forces canary responses to be useless for decrypting records (Lemmas 2 and 3), and in that DKA is a threshold, target-specific and deadline-bound property. Proof-of-custody proposals penalize validators that do not hold attested data [41]. The closest deployed analogue is the tBTC v2 heartbeat, in which each threshold wallet's signing group signs periodic messages and funds leave a wallet that fails one [42]; KEYSTONE instead samples custodians with exact operating characteristics, audits decryption shares under a confidentiality constraint, and models readiness over time. Data-availability proofs and

sampling[9,43] concern public data, which KEYSTONE takes as given. Supplementary Table S3 compares these mechanisms.

*Accountability.* BFT forensics[44] attributes safety violations with transferable evidence, threshold traitor tracing addresses coalitions that leak decryption capability[45], and attributing liveness failures depends on timing[32].

## 10 Threats to validity and limitations

*Novelty.* The components are not new; our claim covers the property decomposition, the confidentiality-preserving audit and its analysis, and the labeled evaluation, and it rests on a bounded literature search.

*Internal validity.* Both designs were hashed before execution, but the hashes are self-recorded and no independent party reviewed the designs; the acceptance-rule and operating-characteristic analyses were added afterwards. R2, R2b and R2c logged fifteen deviations, among them a deadline correction before any testbed run and a restart after the host ran out of ephemeral ports; none changed an endpoint or an analysis in response to outcomes, and the corrected deadline was tested only on fresh R2b traces.

*Construct and external validity.* The audit measures present sampled serviceability, not future cooperation. The temporal model assumes independent exponential failures and repairs with synthetic rates, failure models are synthetic, the testbed uses loopback TCP with synthetic delays, and geo-distributed effects, ledger delay and clock skew are not measured.

*Prototype scope.* A dealer replaces distributed key generation; the contract trusts an administrator for sampling, invalid-response evidence and cancellation; neither code nor contract has been externally audited; the lemmas hold in the random-oracle model with static corruption; and Lemma 4 and the temporal results are given with proof sketches here and full proofs in the supplementary material.

## 11 Conclusion and future directions

Ciphertext availability does not ensure that the threshold service needed for an encrypted-rollup dispute answers in time. We defined dispute-key availability and the weaker properties it is confused with, showed that a beacon-derived canary audit exercises real shares while its transcripts are simulatable from public data, gave its exact operating characteristics, sample sizes and sequential tests that certify a safety margin, a placement condition and a deadline budget, and showed that confidentiality against parties who obtain other releases requires binding releases to records, as record-labeled TDH2 does. For custodians that fail and recover, an exact recursion gives the risk that a dispute fails after any audit history, and a floor shows when auditing more often cannot help. The evaluation confirmed each separation: a small strict audit barely distinguishes a failed committee from a live one and alarms often on healthy committees, selective withholders pass every audit and defeat every dispute, correlated domain failures dominate risk over time, and with elliptic-curve verification off the network loop the network governs dispute completion; an order-statistic deadline held in a replication specified in advance, except without

network delay, where host drift of a few percent defeats any deadline without slack. Audits pay off when they trigger resharing, and a known dispute target can be protected against withholders only by deterrence. Future work should evaluate decoy disputes and stake-based deterrence against selective withholding, which no pre-dispute test detects; readiness models calibrated on operational data; and a production instantiation with distributed key generation and realistic ledger delays, evaluated across regions.

## Statements and Declarations

*Ethical considerations.* This computational study uses synthetic data, simulation, prototype outputs and smart-contract tests, and involves no human participants, personal data or live systems; ethical approval was not required.

*Consent to participate.* Not applicable.

*Consent for publication.* Not applicable.

*Declaration of conflicting interest.* The author(s) declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

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*Data availability.* Code, frozen designs with SHA-256 hashes, deviation logs, raw outputs and the scripts that generate every table, figure and in-text number are provided as anonymized supplementary material. A public repository link is on the title page.

*Declaration of AI use.* AI assistance was used for drafting text, integrating code, scripting experiments and generating figures. The author(s) verified the scientific content, the proofs and every reported number against the generated outputs, and take full responsibility for the article.

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